



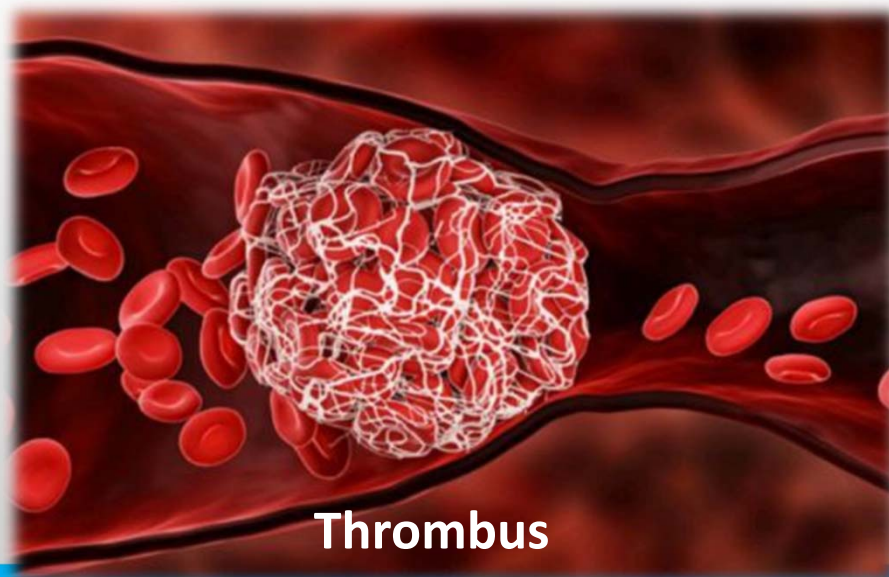
Hemodynamic Disorders

Lec. 8

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Thrombosis

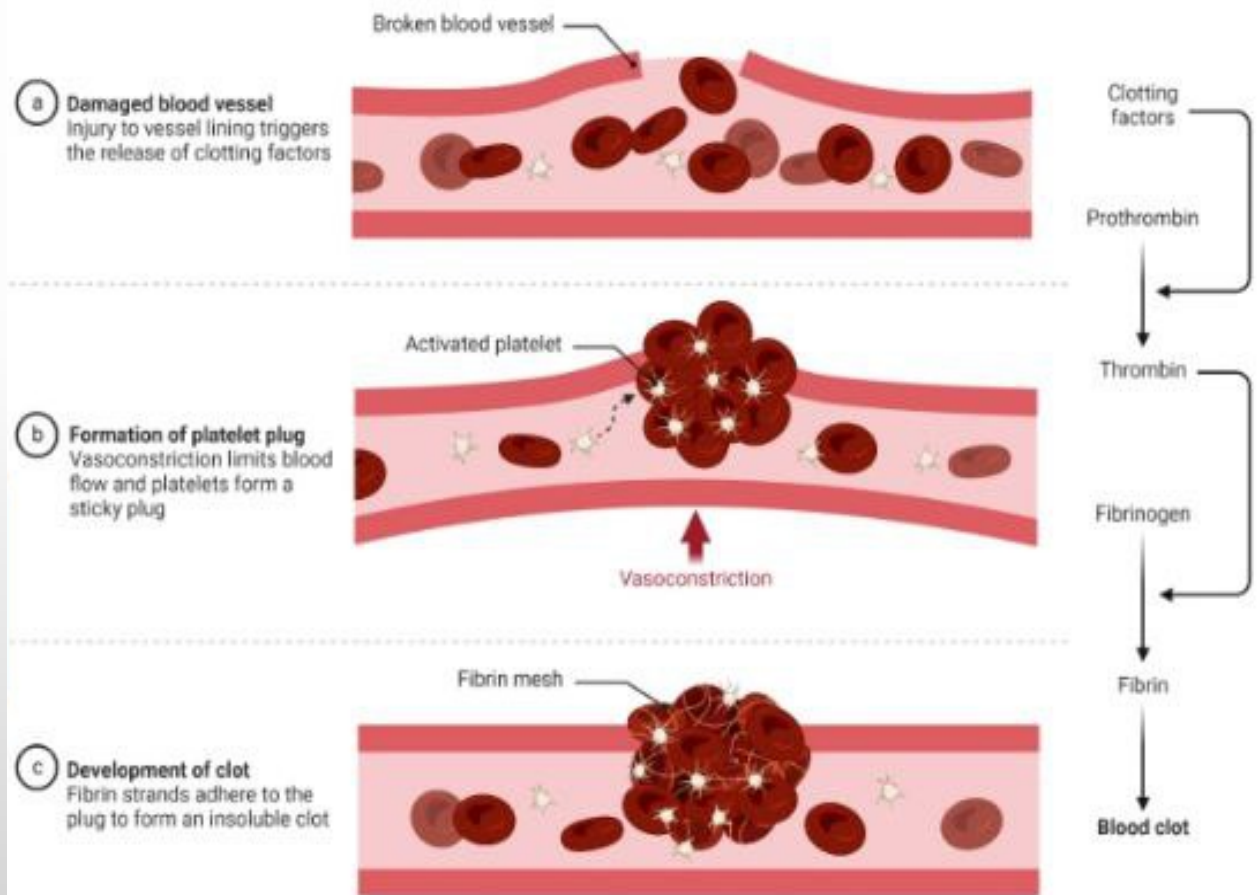
- Is the pathologic counterpart of hemostasis. Thrombosis is the formation of blood clot (thrombus) within non-traumatized, intact vessels.



Thrombus

Hemostasis: Normally the blood is kept in a fluid state with rapid formation of a plug at the site of injury.

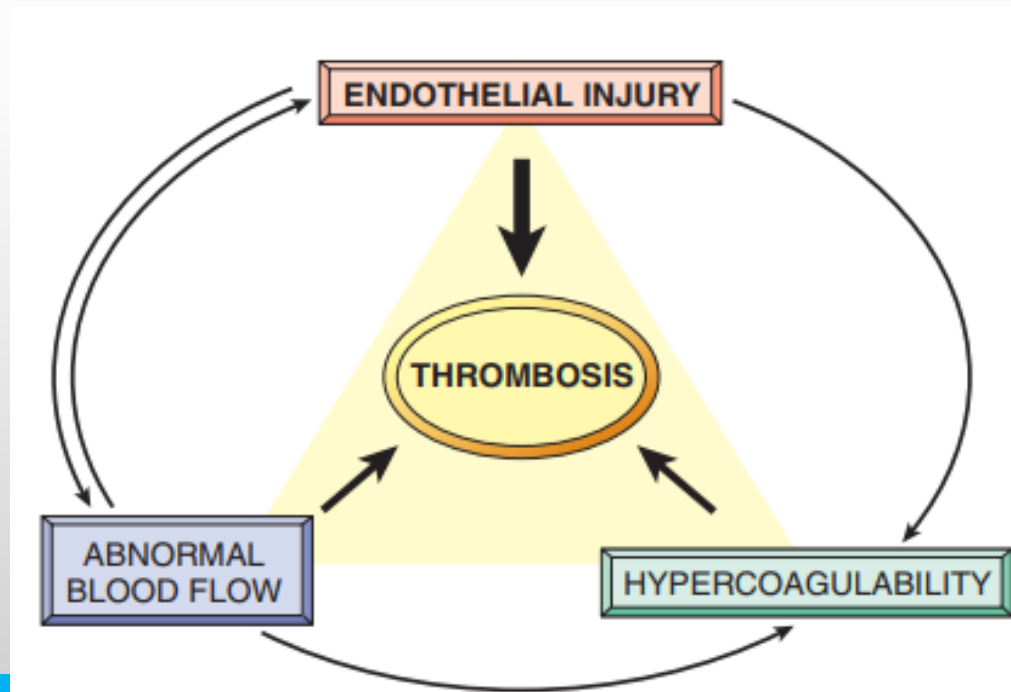
Blood Clot Formation in Broken Vessel



Causes of thrombosis

There are 3 major factors which predispose to thrombosis (Virchow's triad):

1. Endothelial damage.
2. Stasis & turbulence of blood flow.
3. Hypercoagulability .



Fate of the thrombus

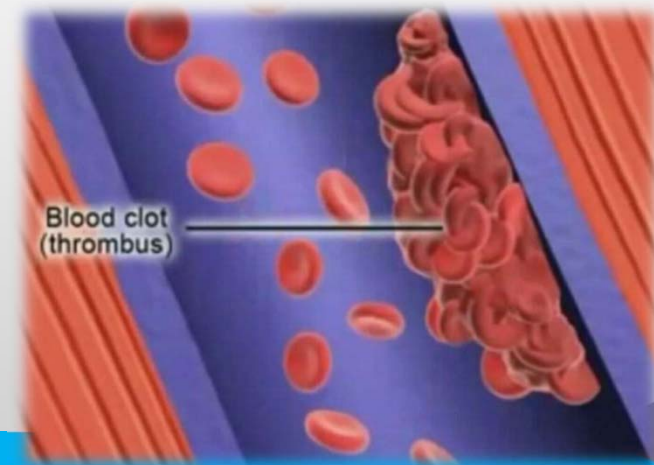
- If a patient survives the initial thrombosis, in the ensuing days or weeks thrombi undergo some combination of the following four events:

1. **Propagation:** The thrombus enlarges through the accretion of additional platelets and fibrin, increasing the odds of vascular occlusion or embolization.

2. **Embolization:** Part or all of the thrombus is dislodged and transported elsewhere in the vasculature.

3. Dissolution: Thrombi are removed by fibrinolytic activity which leads to rapid shrinkage and even total lysis of recent thrombi.

4. Organization and recanalization: Older thrombi become organized by the ingrowth of endothelial cells, smooth muscle cells, and fibroblasts. In time, capillary channels are formed that—to a limited extent—create conduits along the length of the thrombus, thereby reestablishing the continuity of the original lumen

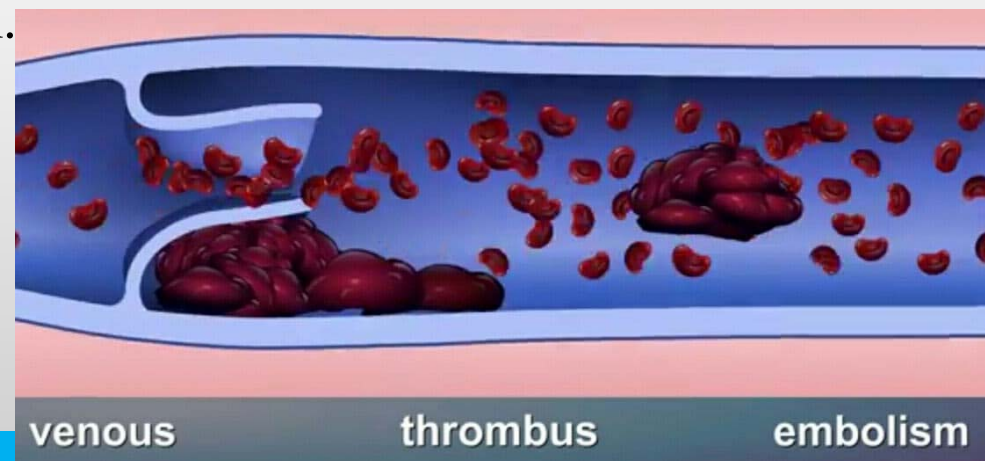


Embolism

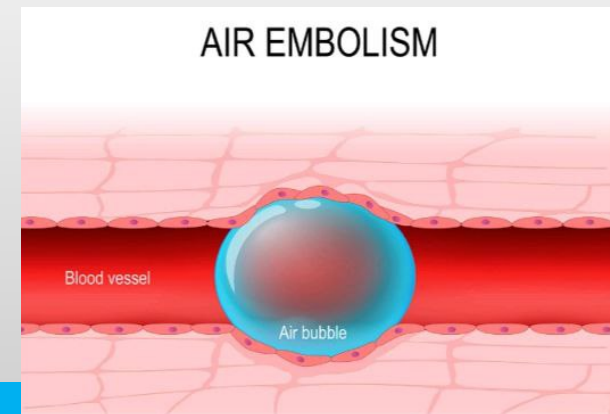
- An embolus is a detached intravascular solid, liquid, or gaseous mass that is carried by the blood from its point of origin to a distant site, where it often causes tissue dysfunction or infarction.

- **Types:**

- 1) The vast majority of emboli derive from a dislodged thrombus, hence the term throm- boembolism.



- Less commonly, emboli are composed of
 - fat droplets
 - bubbles of air or nitrogen
 - atherosclerotic debris (cholesterol emboli)
 - tumor fragments
 - bits of bone marrow
 - amniotic fluid.



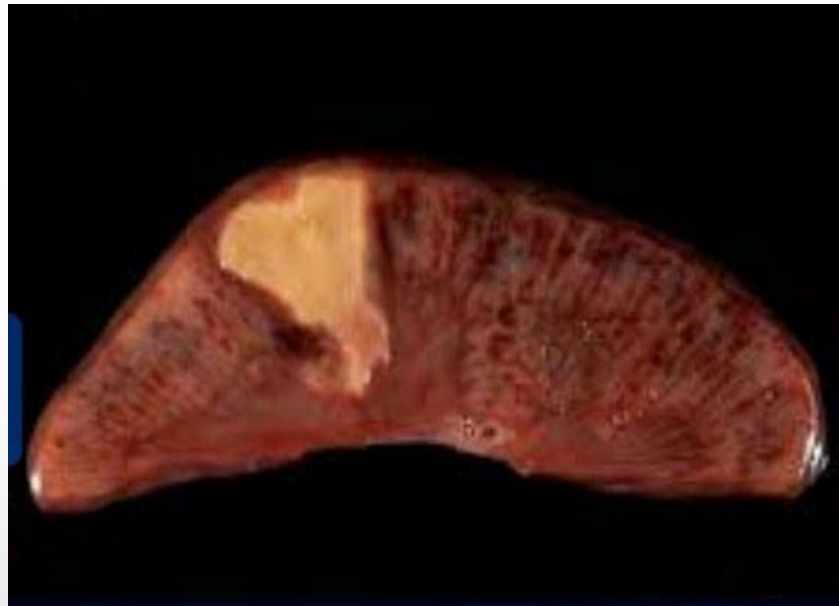


Embolus derived from a lower-extremity deep venous thrombus lodged in a pulmonary artery branch.

- **Pulmonary emboli** derive primarily from lower-extremity deep vein thrombi. Their effects depend mainly on the size of the embolus and the location in which it lodges. Consequences may include right-sided heart failure, pulmonary hemorrhage, pulmonary infarction, or sudden death.
- **Systemic emboli** derive primarily from cardiac mural or valvular thrombi, aortic aneurysms, or atherosclerotic plaques; whether an embolus causes tissue infarction depends on the site of embolization and the presence or absence of collateral circulation.

Infarction

- An infarct is an area of ischemic necrosis caused by occlusion of the vascular supply to the affected tissue.
- Infarcts are classified based on their color (reflecting the amount of hemorrhage) into :
 - 1) **Red (hemorrhagic)** : Infarcts caused by venous occlusion or occurring in loose tissues (e.g., lung).
 - 2) **White (anemic)** occur with arterial occlusions in solid organs with end-arterial circulations (e.g., heart, spleen, and kidney), and where tissue density limits the seepage of blood from adjoining patent vascular beds.



Gross pathology of a kidney section with a zone of pale infarction.



Red Infarction in the lung

Factors that influence development of an infarct

- 1) **Anatomy of the vascular supply:** The presence or absence of an alternative blood supply is the most important factor in determining whether occlusion of an individual vessel causes damage.
- 2) **Rate of occlusion:** Slowly developing occlusions are less likely to cause infarction because they allow time for the development of collateral blood supplies.

3) **Tissue vulnerability to hypoxia:** Neurons undergo irreversible damage when deprived of their blood supply for only 3 to 4 minutes. Myocardial cells die after only 20 to 30 minutes of ischemia. By contrast, fibroblasts within myocardium remain viable after many hours of ischemia.

Shock

- Shock is defined as a state of systemic tissue hypoperfusion resulting from reduced cardiac output and/or reduced effective circulating blood volume .Shock can lead to hypoxic tissue injury if not corrected.
- **The major types of shock are:**
 - Cardiogenic shock : Results from failure of the cardiac pump (e.g., myocardial infarction)
 - Hypovolemic shock : Results from loss of blood (e.g., hemorrhage)
 - Septic shock : is caused by the host response to bacterial or fungal infections

